

Software Engineer

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Experience

SDE 2
VideoVerse

Remote
Jan 2026 – present

- Designed and built **SmartThumbnails** from scratch in **Python/FastAPI** — reverse-engineered selection heuristics from client-published thumbnails into a layered pipeline: **PySceneDetect** scene segmentation, Laplacian sharpness filtering, and **DSPy**-structured Gemini for final selection. Reached **99.2% client acceptance** across the first 4,500 clips, replacing manual curation.
- Built an **SNS-driven** clip enrichment pipeline issuing **DSPy**-structured Gemini calls, so every clip is searchable and tagged at ingest.
- Extended the export pipeline with multi-language, EPG, and AI tagging by routing **SNS events** through the internal AI Gateway, letting broadcast clients localize content without manual work.
- Engineered a hybrid algorithmic + LLM pipeline for sibling-clip metadata back-fill and conflict detection, with batched updates that respect hard SLA windows — cut **SLA breaches by 25%**.
- Unified two fragmented broadcast scheduling sources into a single queryable API; reconciliation logic catches shows split across source APIs that the prior manual-import workflow dropped.
- Owned both the cross-service instrumentation refactor and the collector pipeline for the **Datadog → OpenTelemetry** migration, giving the team vendor-neutral structured observability.

SDE 1
VideoVerse

Remote
Nov 2024 – Jan 2026

- Built an internal LLM evaluation harness from scratch — eval datasets, metric definitions, and **Langfuse** integration — and used it to drive ground-truth analysis that lifted output accuracy **from 68% to 92%**.
- Tuned and applied **DSPy MIPRO/COPRO** and **GEPA** prompt optimizers per service for thumbnail selection and metadata generation, raising task-level accuracy **from 30% to 87%**.
- Built a **FastAPI AI Gateway** fronting 5 backend AI services with service-to-service token auth and retry/backoff baked into both FastAPI middleware and the **HTTPX** transport — **zero dropped jobs** across 50k+ clips and 300+ streams monthly.
- Engineered a **DSPy + Gemini** clip-ratings pipeline running on **500k+ clips/month**, with rate-limit-aware scheduling so peak-load batches still complete before SLA breach.
- Established Langfuse and OTEL alerting workflows to enforce SLA compliance across the AI services.
- Co-led migration of AWS infrastructure to **Terraform**, extending the existing module structure and adding **OTEL** wiring plus a **Cloudflare Tunnel** for secure local access to internal services.

Software Engineer Intern
VideoVerse

Remote
June 2024 – Nov 2024

- Originally interned at **LetsDive.io** as a Django backend engineer; continued the role at VideoVerse **following its acquisition of LetsDive.io**.
- Designed an ETL feed from **PostgreSQL and MongoDB** into **BigQuery** with **sub-100ms** end-to-end latency, since live frontend and backend services consume from BigQuery in the request path.
- Rewrote the legacy thumbnail service storage layer from **SQLite to PostgreSQL** with a from-scratch schema (no back-compat), and built the face-embedding ingestion and deployment that drives selection accuracy.

Projects

Battlefield Portal Library | [bfportal.gg](#) | <https://gh.bfportal.gg>

Jan 2022 – present

- Django/Jinja** CMS, currently migrating templates to **HTMX** for dynamic interactions without a JavaScript framework; **Cloudflare** edge caching, Docker containerization, and GitHub Actions CI/CD.

Gametools Network | [gametools.network](#) | [github.com/community-network](#)

May 2022 – present

- Open-source Stats API behind **Envoy** routing into a **Kubernetes** cluster, serving **2M+ daily requests** extended the existing Envoy logic and authored the Kubernetes autoscaling policies for upstream services.

Skills

AI / ML: DSPy, MIPRO/COPRO, GEPA, Langfuse, Google Gemini, VertexAI, Prompt Engineering, LLM Evaluation & Optimization

Backend: Python, Go, TypeScript, FastAPI, Django, PostgreSQL, MongoDB, Redis, BigQuery

Infrastructure: AWS (SNS, ECS), Terraform, Docker, Kubernetes, OpenTelemetry, Grafana, Envoy, Cloudflare, GitHub Actions

Education

MIET
B.Tech in Computer Science Engineering, Minor in Data Science

Meerut, Uttar Pradesh
Jan 2020 – Jan 2024